



Leibnitz Theorem



1) If $y = \frac{\log x}{x}$ then P.T $y_5 = \frac{5!}{x^6} [1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} - \log x]$

$$\text{Let } u = \frac{1}{x} = x^{-1}$$

$$v = \log x$$

$$u_1 = (-1)x^{-2} = \frac{-1}{x^2}$$

$$v_1 = \frac{1}{x}$$

$$u_2 = (-1)(-2)x^{-3} = \frac{2!}{x^3}$$

$$v_2 = \frac{-1}{x^2}$$

$$u_3 = (-1)(-2)(-3)x^{-4} = \frac{-3!}{x^4}$$

$$v_3 = \frac{2!}{x^3}$$

$$u_4 = (-1)(-2)(-3)(-4)x^{-5} = \frac{4!}{x^5}$$

$$v_4 = \frac{-3!}{x^4}$$

$$u_5 = (-1)(-2)(-3)(-4)(-5)x^{-6} = \frac{-5!}{x^6}$$

$$v_5 = \frac{4!}{x^5}$$

$$y_5 = (uv)_5 = u_5 v + 5u_4 v_1 + \frac{5.4}{2!} u_3 v_2 + \frac{5.4.3}{3!} u_2 v_3 + \frac{5.4.3.2}{4!} u_1 v_4 + uv_5.$$

$$= \frac{-5!}{x^6} \log x + 5 \frac{4!}{x^5} \frac{1}{x} + \frac{5.4}{2} \left(\frac{-3!}{x^4} \right) \left(\frac{-1}{x^2} \right) + \frac{5.4.3}{3!} \left(\frac{2!}{x^3} \right) \left(\frac{2!}{x^3} \right) + \frac{5.4.3.2}{4!} \left(\frac{-1}{x^2} \right) \left(\frac{-3!}{x^4} \right) + \frac{1}{x} \frac{4!}{x^5}$$

$$= \frac{5!}{x^6} \left\{ -\log x + 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} \right\} = \frac{5!}{x^6} \left[1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} - \log x \right]$$



2) If $y = x^2 \cos 3x$ then find y_n .

3) If $y = 5^x x^5$ then find y_5 .



4) If $y = e^{\tan^{-1} x}$ then P.T $(1+x^2)y_{n+1} + (2nx - 1)y_n + n(n-1)y_{n-1} = 0$

$$y = e^{\tan^{-1} x}$$

$$y_1 = e^{\tan^{-1} x} \frac{1}{1+x^2}$$

$$(1+x^2)y_1 = e^{\tan^{-1} x}$$

$$(1+x^2)y_1 = y \dots\dots 1$$

Apply Leibnitz theorem

$$[(1+x^2)y_1]_n = [y]_n$$

$$u = y_1 \quad v = (1+x^2)$$

$$u_1 = y_2 \quad v_1 = 2x$$

$$u_2 = y_3 \quad v_2 = 2$$

$$u_3 = y_4 \quad v_3 = 0$$

$$\cdot \quad \cdot$$

$$\cdot \quad \cdot$$

$$u_{n-1} = y_n \quad v_{n-1} = 0$$

$$u_n = y_{n+1}$$

$$v_n = 0$$

$$(uv)_n = u_n v + n u_{n-1} v_1 + \dots + \frac{n(n-1)}{2!} u_{n-2} v_2 + \frac{n(n-1)(n-2)}{3!} u_{n-3} v_3 + \dots$$

$$= (1+x^2)y_{n+1} + n y_n 2x + \frac{n(n-1)}{2!} y_{n-1} 2 + 0$$

$$(1+x^2)y_{n+1} + n y_n 2x + n(n-1) y_{n-1} = y_n$$

$$(1+x^2)y_{n+1} + (2nx - 1)y_n + n(n-1)y_{n-1} = 0$$



5) If $y = e^{m \sin^{-1} x}$ then P.T $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2+m^2)y_n = 0$.

$$y = e^{m \sin^{-1} x}$$

$$y_1 = e^{m \sin^{-1} x} m \left(\frac{1}{\sqrt{1-x^2}} \right)$$

$$\sqrt{1-x^2} y_1 = my \dots\dots 1$$

$$\sqrt{1-x^2} y_2 + \frac{-2x}{2\sqrt{1-x^2}} y_1 = my_1$$

Multiplying by $\sqrt{1-x^2}$

$$(1-x^2)y_2 - xy_1 = m\sqrt{1-x^2}y_1$$

$$(1-x^2)y_2 - xy_1 = m^2y$$

$$(1-x^2)y_2 - xy_1 - m^2y = 0 \dots 2$$

Differentiating eq. 2 n times by using Leibnitz thm, we get

$$\{(1-x^2)y_2\}_n - \{xy_1\}_n - \{m^2y\}_n = 0 \dots 2$$

$$(1-x^2)y_{n+2} + ny_{n+1}(-2x) + \frac{n(n-1)}{2!}y_n(-2) - \{xy_{n+1} + n(1)y_n\} - m^2y_n = 0$$

$$(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2+m^2)y_n = 0.$$



- If $x = \tan \log y$ then P.T $(1+x^2)y_{n+1} - (2nx-1)y_n + n(n-1)y_{n-1} = 0$.